

African Climate Trend Analysis (Week 0 Challenge)

10 Academy – AI Mastery Program

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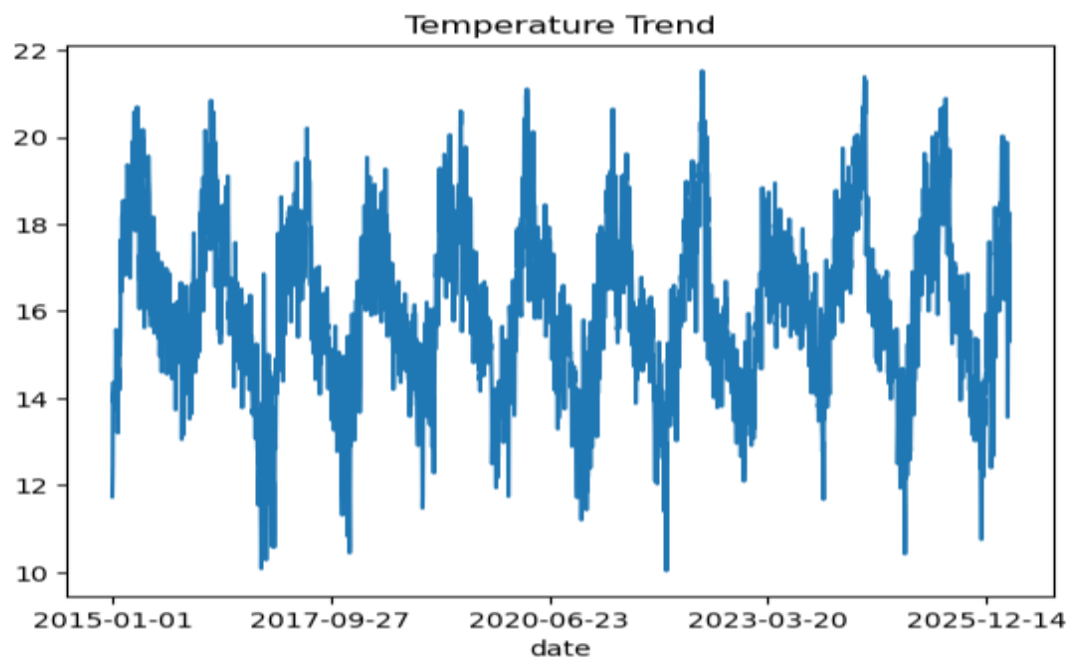
Introduction

This report presents an exploratory analysis of historical climate data for Ethiopia and selected African countries using NASA POWER dataset (2015–2026). The goal is to identify climate trends, seasonal patterns, and extreme events to support evidence-based climate discussions ahead of COP32.

Data Overview

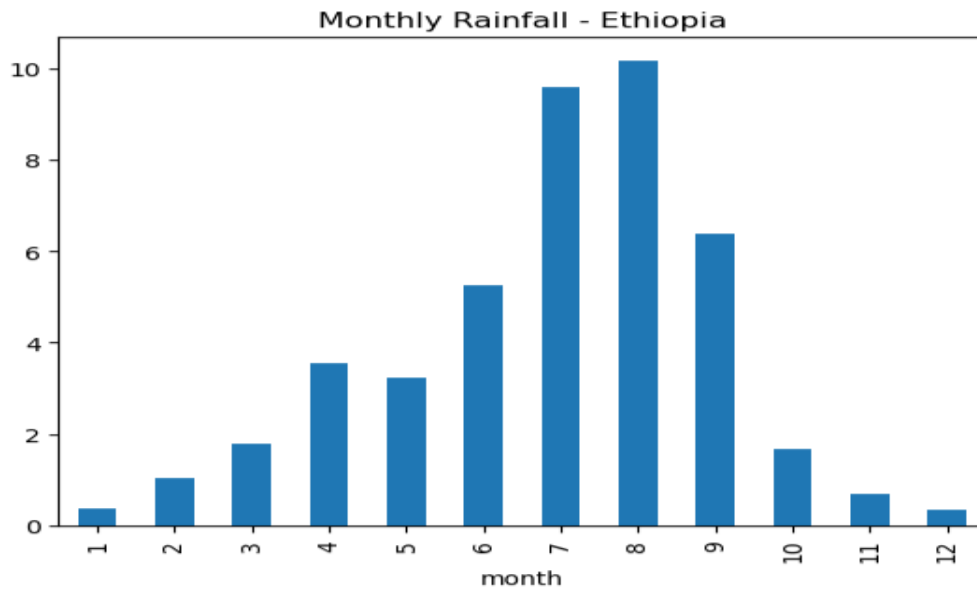
The dataset contains daily meteorological variables including temperature (T2M, T2M_MAX, T2M_MIN), precipitation (PRECTOTCORR), humidity (RH2M), wind speed (WS2M), and atmospheric pressure (PS). Missing values encoded as -999 were cleaned and replaced with NaN.

Key Charts Section (VERY IMPORTANT)



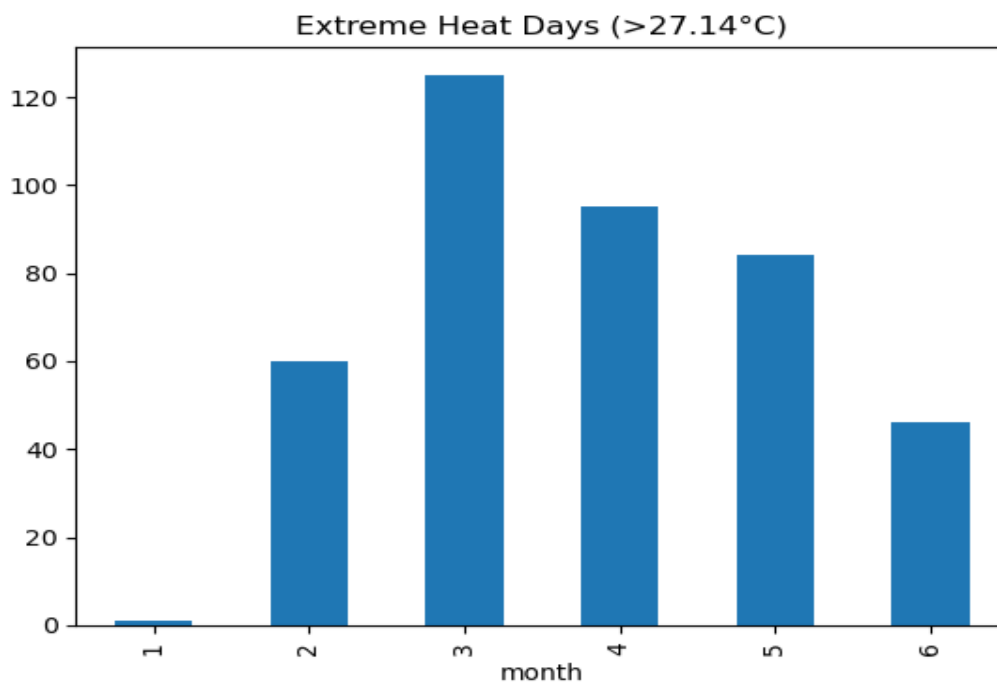
Temperature shows clear seasonal variation with consistent fluctuations across the study period.

Rainfall Pattern



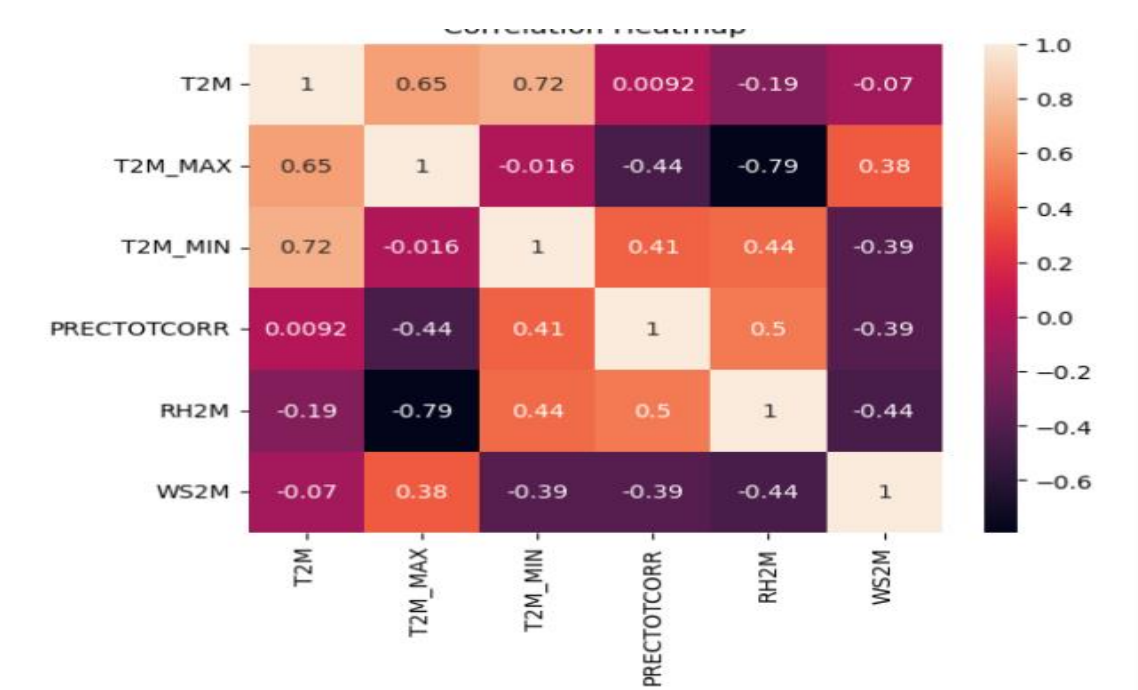
Precipitation shows strong seasonality, indicating distinct wet and dry periods.

Extreme Heat



Extreme temperature events were defined using a data-driven threshold (90th percentile). Results show moderate occurrence of heat extremes.

Correlation Heatmap



Temperature variables are strongly correlated, while precipitation shows weaker correlation with other meteorological factors.

COP32 Insight

- Temperature patterns show clear seasonal variability, indicating climate stability with periodic warming phases.
- Precipitation variability suggests strong seasonal dependence, important for agriculture planning.
- Extreme heat events, while limited, show potential risk under climate change scenarios.
- Ethiopia’s climate profile is comparable to regional patterns, showing no extreme deviation from neighboring countries.
- Data supports the need for continued investment in climate monitoring and adaptation systems.

Conclusion

This analysis provides a foundational understanding of climate behavior in Ethiopia and surrounding regions. The findings highlight seasonal patterns and moderate extreme event risks, supporting the need for data-driven climate resilience strategies.